Felix Hiller

Personal Data

Date of Birth:	16 February 1996
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Work Experience

October 2023 - Present *Software Engineer* **Foundry**
Developer working on Nuke, working on:

- Multishot features; including GSV subsystem, Group View and user callbacks.
- Leading the integration of Griptape Nodes into Nuke
- Implementation of MaterialX into Nuke's latest 3D system

Working with C++20, Qt, Python, Git and Conan

November 2022 - October 2023 *Software Engineer* **ETAS**
Developer for the Automated Driving Alliance, working on embedded middleware components and architecture to enable scaling of a self-driving car system. Driving the refactor of a maturing codebase to help test and reach satisfactory safety standards.

Working with C++17, Python, Git and Conan

March 2019 - May 2022 *Software Engineer* **Autodesk**
Developer on Fusion 360 and Powermill, working primarily on:

- Implementation of novel sweep-line algorithm to improve triangulation reliability
- Construction of Toolpath creation algorithms, helped improve stability of new corner finishing strategies
- Responsible for kernel integration, including the design and architecture of Toolpath Edits within Fusion
- Instigated large refactoring and overhaul of critical underlying systems to better adhere to C++17 standards and design paradigms

Working with C++17, Python, Typescript, Git.

Education

2014 - 2018 *First class Masters Degree in Computer Science* University of Birmingham
2012 - 2014 *A-Levels* Hills Road Sixth Form College, Cambridge
2007 - 2012 *Eight GCSEs* Coleridge Community College, Cambridge

Technical skills

Programming Languages:
C, C++: 7 years of professional experience
Python: 3 years of professional experience

Projects

C++ **Raytracer with novel non-uniform octree optimisation** GitHub
A Novel ray-tracing optimisation, using non-uniform octrees to subdivide the scene more tightly around a given model. Produces images from x3d files and compares rendering times with and without the optimisation.

C++ **Implementation of board-game Azul: Summer Pavilion** GitHub
Created using SFML, including design of AI system to replicate complex human-like strategies in Azul's varied and complex decision space.

Java **Super Noughts and Crosses** GitHub
Noughts and crosses played on a 3x3 grid of 3x3 grids. Implemented a strong AI using traditional alpha beta pruning methods.

Interests and Hobbies

Avid climber and mountaineer
Voracious board game and puzzle enthusiast